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Bipod

Abstract

Interface connection adapter arrangement (300) for a bipod (1) for a rifle (2), wherein the arrangement is arranged to be pivotally connected to two leg parts (110), the leg parts when fastened both being arranged to pivot in relation to the arrangement between a first orientation in which an axis (111) of the leg part in question extends on a first side of a main connection plane (303), and a second orientation in which the axis extends on a second, opposite, side of said main connection plane. The invention is characterised in that the arrangement further comprises a first interface side (330), provided with a first rifle fastener (331), in turn arranged to fasten the arrangement to the rifle in accordance with a first rifle fastening system, and a second interface side (340), provided with a second rifle fastener (341), in turn arranged to fasten the arrangement to the rifle in accordance with a second rifle fastening system, and in that the second rifle fastening system is associated with a different fastener geometry as compared to the first rifle connection system. The invention also relates to a method.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4288939	12/1980	Beretta	42/94	F41A 23/08
5794899	12/1997	Tamlllos	N/A	N/A
6829857	12/2003	Houtsma	89/37.04	F41A 23/10
8402684	12/2012	Beltz	42/72	F41A 23/10
8806796	12/2013	Clifton	N/A	N/A
2006/0278797	12/2005	Keng	N/A	N/A
2008/0006747	12/2007	Bobro	248/178.1	F41A 23/10
2010/0218410	12/2009	Cabahug et al.	N/A	N/A
2010/0218411	12/2009	Keng	42/90	F41A 23/10
2010/0218670	12/2009	Keng	N/A	N/A
2012/0144717	12/2011	Peterson	42/90	F41G 11/003
2012/0273631	12/2011	Deros	N/A	N/A
2019/0072355	12/2018	Pop	N/A	F41A 23/10
2019/0145731	12/2018	Stewart	42/90	F41C 27/00
2019/0162362	12/2018	Chen	N/A	F16B 2/185
2019/0331460	12/2018	Johnson, Sr.	N/A	N/A
2021/0102783	12/2020	Ma et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020214803	12/2019	WO	N/A

OTHER PUBLICATIONS

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Background/Summary

- (1) The present invention relates to a bipod, and in particular to a bipod for a rifle, such as a firearm or an air rifle. The invention also relates to a method of operating such a bipod.
- (2) In particular aspects of the invention, as will be understood from the following, the invention relates to a detachable leg arrangement for use as a part of such a bipod; an adjustable leg fastening arrangement for use as a part of such a bipod; and an interface connection adapter arrangement for use as a part of such a bipod.
- (3) Bipods are used for competitive and recreational shooting, as well as for hunting. When supported using a bipod, a firearm or other rifle can be oriented with a particular desired adjustable aim and be held steadily. Such a bipod may be arranged as an integrated part of the rifle in question, but more usually is fastened to the rifle using any one of a number of available fastening systems, such as the Picatinny and ARCA systems. Such fastening system may offer an adjustable fastening point for the bipod along a length of the rifle. Moreover, a height and/or angle of the support provided by the bipod may be adjustable before and/or during use. See, for instance, <https://en.wikipedia.org/wiki/Bipod> and pages linked therefrom for information about various conventional rifle fastening systems.
- (4) In particular, a desired orientation of the rifle may be achieved by adjusting the length of the legs of the bipods, such as via said legs being telescopic to adjust said length.
- (5) There are a number of desired properties of such bipods.
- (6) Hence, they should preferably offer a very stable yet finely adjustable orientation of the rifle. Such adjusting should be easily imparted by a user, preferably using a single hand, quickly and reliably. They should also be robust and withstand the often harsh conditions in the field, during transport and use. They should be flexibly usable in many different situations and with different rifles. Furthermore, they should offer full safety for the user, while still not being too complex so as to allow inexpensive manufacturing.
- (7) Some known bipods include those marketed by Atlas Bipods (see <https://www.accushot.com/AtlasBipods.php>), Harris Bipods (see <https://www.harrisbipods.com>) and Accutac (see <https://www.accu-tac.com>).
- (8) Known bipods may offer one or several of said desired properties, but are typically associated with design compromises.
- (9) The present invention solves the above described problems, presenting a modular bipod where each of the modules in themselves, and the assembled bipod as a whole, has one or more of said properties.
- (10) Hence, the invention relates to an interface connection adapter arrangement for a bipod for a rifle, the interface connection adapter arrangement being associated with a main connection plane parallel to both a longitudinal direction and a lateral direction, said longitudinal direction being perpendicular to said lateral direction, wherein the interface connection adapter arrangement is

arranged to be pivotally connected to two leg parts, a respective pivot point of both leg parts being arranged in said main connection plane, the leg parts when thus fastened both being arranged to pivot in relation to the interface connection adapter arrangement between a first orientation in which an axis of the leg part in question between said pivot point in question and a distal end of the leg part in question extends on a first side of said main connection plane, and a second orientation in which an axis of the leg part in question between said pivot point in question and a distal end of the leg part in question extends on a second, opposite, side of said main connection plane, the interface connection adapter arrangement being characterised in that the interface connection adapter arrangement further comprises a first interface side, arranged on said first side of the main connection plane, provided with a first rifle fastener, the first rifle fastener being arranged to fasten the interface connection adapter arrangement to the rifle in accordance with a first rifle fastening system, and a second interface side, arranged on said second side of the main connection plane, provided with a second rifle fastener, the second rifle fastener being arranged to fasten the interface connection adapter arrangement to the rifle in accordance with a second rifle fastening system, and in that the second rifle fastening system is associated with a different fastener geometry as compared to the first rifle connection system.

(11) The invention further relates to a method for operating a bipod comprising an interface connection adapter arrangement of said type, the method comprising the steps a) providing said interface connection adapter arrangement, being attached to each of said leg parts in a respective orientation in which an axis of the leg part in question between the pivot point of the leg part in question and a distal end of the leg part in question extends on a first side of said main connection plane; b) connecting the interface connection adapter arrangement to a rifle using said first rifle fastener; c) detaching said first rifle fastener; d) pivoting each of said leg parts to a respective orientation in which an axis of the leg part in question between the pivot point of the leg part in question and a distal end of the leg part in question extends on a second side of said main connection plane; and e) connecting the interface connection adapter arrangement to a rifle using said second rifle fastener.

Description

(1) In the following, the invention will be described in detail, with reference to exemplifying embodiments of the invention and to the enclosed drawings, wherein:

(2) FIG. 1 is a perspective view of an exemplifying bipod according to the present invention, together with a mounted rifle and with a leg part of the bipod in a first angular orientation;

(3) FIG. 2 is a perspective view of the bipod illustrated in FIG. 1, but without said rifle and with said leg part in a second angular orientation;

(4) FIG. 3 is an exploded perspective view of the bipod;

(5) FIG. 4 is a partly removed perspective view of the bipod, showing a leg part, a detachable leg arrangement and an adjustable leg fastening arrangement of said bipod;

(6) FIG. 5 is a perspective detail view of said detachable leg arrangement, showing the leg part, a leg part attachment means and a leg fastening part of said bipod;

(7) FIG. 6 is a perspective detail view of said detachable leg arrangement, showing the leg part attachment means and the leg fastening part;

(8) FIG. 7 is a perspective detail view of said detachable leg arrangement, showing the leg fastening part;

(9) FIG. 8 is a perspective detail view of said detachable leg arrangement, showing the leg part attachment means;

(10) FIG. 9 is a partly removed perspective view of said bipod, showing the detachable leg arrangement, the adjustable leg fastening arrangement and an interface connection adapter

arrangement of said bipod;

(11) FIGS. **10a** and **10b** are respective exploded perspective views of said adjustable leg fastening arrangement;

(12) FIG. **11** is a perspective view of said adjustable leg fastening arrangement;

(13) FIG. **12** is a partly removed perspective view of said interface connection adapter arrangement;

(14) FIG. **13** is a perspective detail view of a freedom of movement adjustment means of said bipod;

(15) FIG. **14** is a flow chart illustrating a method according to a detachable leg attachment aspect of the present invention;

(16) FIG. **15** is a flow chart illustrating a method according to an adjustable leg fastening aspect of the present invention;

(17) FIG. **16** is a flow chart illustrating a method according to a reversible interface connection aspect of the present invention;

(18) FIG. **17** is a flow chart illustrating a method according to a quick-release interface connection aspect of the present invention;

(19) FIG. **18** is an exploded perspective view illustrating a leg locking mechanism according to an embodiment of the invention;

(20) FIG. **19** is a perspective cross-sectional view illustrating said leg locking mechanism; and

(21) FIG. **20** is a partly removed perspective view illustrating said leg locking mechanism.

(22) It is noted that all FIGS. **1-13** share the same reference numerals, and illustrate one and the same bipod **1** according to the invention. However, it is clear that the illustrated bipod **1** is only an embodiment example, selected in order to exemplify various details of the different aspects of the present invention.

(23) FIGS. **18-20** illustrate an alternative or supplementary embodiment, but also shares the same reference numerals, for same or corresponding parts, as used in FIGS. **1-13**.

(24) It is further noted that FIGS. **1-13** and **18-20** are somewhat simplified, and that for instance threads are only shown symbolically rather than in accurate detail.

(25) Hence, as is illustrated in the Figures, a bipod **1** according to the invention is a bipod **1** for a rifle **2**, in the sense that the bipod **1** is arranged for use with the rifle **2** to support the rifle **2** on a support of some sort, such as on a table or on the ground, during use of the rifle **2** for shooting.

(26) The rifle **2** may be a firearm, such as a gunpowder rifle; an air rifle; or any other type of rifle.

(27) The bipod **1** may be a detachable bipod **1**, arranged to be releasably fastened to the rifle **2** using a mechanical connection interface, as will be discussed below.

(28) The bipod **1** comprises two leg parts **110**, each such leg part **110** having an axial direction **111**, an axially distal end **112** and an axially proximal end **113**. The axially distal end **112** may comprise a foot part **117**, which may be of plastic material, rubber, metal material or any other suitable material; and may be formed as a surface friction foot arranged to achieve high friction between the foot part **117** and a support surface; a pointy foot arranged to protrude down into a soft support surface; or in any other per se conventional, suitable way so as to provide a stable, reliable support for the rifle **2**.

(29) The foot part **117** may be replaceable, so that the leg part **110** can be used with several different types of foot parts, and/or so that the bipod **1** may be arranged with replaceable leg parts **110** so as to cater to different such needs, as the case may be and as discussed below.

(30) Each leg part **110** may be made from metal material, such as aluminium or steel, or in any other suitable material, such as a carbon fibre-containing material or a suitable thermoplastic. It may be axially **111** extendible, such as telescopingly extendible. However, in some embodiments each leg part **110** is non-extendible and stiff between said distal end **112** and said proximal end **113**. The leg part **110** may even be made from a single, integrated piece of material extending all the way between said respective extreme ends **112**, **113**, and in particular between the foot part **117** and

a fastening point (such as the below-discussed first threaded engagement means **114**) for fastening the leg part **110** to the rest of the bipod **1**, such as using a leg part attachment means **120** of the below-discussed type.

(31) As used herein, the term “integrated piece of material” means a piece of material without any joints, such as a single connected and/or homogenous piece of cast and/or machined metal. Such an integrated piece of material may be hollow or solid, depending on a desired weight, stability or ease of transport.

(32) Namely, according to said detachable leg attachment aspect of the present invention, the bipod **1** further comprises such a leg part attachment means **120**, arranged to fasten the leg part **110** to the rest of the bipod **1**. The leg part attachment means **120**, in turn, comprises a second threaded engagement means **121**.

(33) Furthermore according to said detachable leg attachment aspect, said proximal end **113** comprises said first threaded engagement means **114**. This first threaded engagement means **114** is arranged to, when the leg part attachment means **120** is in an engaged state, threadedly engage with said second engagement means **121** so as to allow the leg part attachment means **120** to move axially towards the leg part **110** by moving the leg part attachment means **120** into deeper engagement with the leg part **110**.

(34) Said engaged state of the leg part attachment means **120** is illustrated in FIGS. **4** and **5**, and involves the leg part attachment means **120** engaging with the leg part **110** by use of said threaded engagement means **114**, **121**, such as external threads of the leg part attachment means **120** threadedly engaging with corresponding internal threads of the leg part **110**. The engagement may be axial **111**, in the sense that the leg part attachment means **120** may move axially **111** in relation to the leg part **110** as the threaded engagement is deepened or loosened.

(35) The leg part attachment means **120** may be a separate part, arranged to engage both with the leg part **110** and a leg fastening part **130** of the type discussed below.

(36) According to said adjustable leg fastening aspect of the present invention, the bipod **1** comprises an adjustable leg fastening arrangement **200** for the bipod **1**, for adjusting an orientation of the or each leg part(s) **110** in relation to the rest of the bipod **1** and as a result in relation to the rifle **2** to which the bipod **1** is attached.

(37) The adjustable leg fastening arrangement **200** according to this adjustable leg fastening aspect comprises one or two first pivoting parts **210**, each being arranged to be fastened to a respective one of the leg parts **110**. The adjustable leg fastening arrangement **200** also comprises one or two second pivoting parts **220**, each being arranged to be fastened to a bipod **1** main part, or directly to the rifle **2**. The bipod **1** main part may, for instance, be the interface connection adapter arrangement **300** discussed below.

(38) Then, each of said second pivoting parts **220** is pivotable in relation to a corresponding one of the one or two first pivoting parts **210**, so that the corresponding leg part **110** as a result is pivotable in relation to the bipod **1** main part and as a result to the rifle **2**, or directly to the rifle **2**, as the case may be.

(39) Namely, according to both said reversible interface connection aspect of the present invention and according to said quick-release interface connection aspect of the invention, the bipod **1** comprises an interface connection adapter arrangement **300** for the bipod **1**, arranged to fasten the bipod **1** to the rifle **2** using a mechanical connection interface, in the sense that an interface part of the rifle **2** mechanically engages with the interface connection adapter arrangement **300** so as to detachably connect the bipod **1** to the rifle **2** in a stable enough manner so that the bipod **1** can be used as a stable and robust support for the rifle **2** during use of the rifle **2**.

(40) The interface connection adapter arrangement **300** is associated with a main connection plane **303**, being parallel to both a longitudinal direction **301** and a lateral direction **302** of the interface connection adapter arrangement **300**. The longitudinal direction **301**, which may be parallel to a main longitudinal sliding direction of a rifle **2** interface part and/or a main longitudinal or shooting

direction of the attached rifle **2**, is perpendicular to said lateral direction **302**. When the attached rifle **2** is in a neutral shooting orientation, for horizontal shooting, the lateral direction **302** may be horizontally aligned.

(41) When the rifle **2** is oriented in such a normal shooting orientation, a shooting direction of the rifle being horizontal, it is furthermore preferred that the main connection plane **303** is also horizontal.

(42) Further according to said reversible interface connection aspect, the interface connection adapter arrangement **300** is arranged to be pivotally connected to said two leg parts **110**, a respective pivot point (centre of pivoting or rotation) of each of the leg parts **110** being arranged in said main connection plane **303**.

(43) This pivotal connection then allows each of the leg parts **110**, when thus fastened, to pivot in relation to the interface connection adapter arrangement **300** between a first (pivot) orientation and a second (pivot) orientation.

(44) In the first orientation, an axial direction **111** axis of the leg part **110** in question between said pivot point in question and the distal end **112** of the leg part **110** in question extends on a first side of the main connection plane **303**. In the second orientation, said axis of the leg part **110** in question extends on a second, opposite, side of said main connection plane **303**. The first orientation may be 180° apart from the second orientation.

(45) In other words, according to said reversible interface connection aspect, each of the leg parts **110** can be pivoted (swung) so as to extend on either side of the main connection plane **303**. This may imply that the leg part **110** in question may be pivoted so that a majority of the leg part **110** in question extends, in a direction perpendicular to said main connection plane **303**, on a first interface side **330** (see FIG. 12) when in said first orientation, and on a second interface side **340** when in said second orientation.

(46) The pivoting of each leg part **110** typically takes place about the lateral direction **302**, or at least about a direction not parallel to the longitudinal direction **301**, such as parallel to the main connection plane **303**. Each of the leg parts **110** may even be pivotal over a full 360° interval in relation to this pivoting axis about which the pivoting takes place.

(47) The pivoting of each leg part **110** achieves that a direction of the leg part **110** in question, in relation to an attached rifle **2**, can be adjusted according to need. For instance, in order to lower or raise the rifle **2**, or adjust a shooting angle of the rifle **2** in relation to a support surface, both leg parts **110** can be pivoted forwards or backwards so as to adjust an angle of the leg part **110** in question in relation to a main shooting direction of the rifle **2**. In order to tilt the rifle **2**, one leg part **110** may be pivoted in relation to the other leg part **110**.

(48) Comparing FIGS. 1 and 2, FIG. 2 illustrates the bipod **1** having the leg part **110** closest to the viewer pivoted to a different pivot angle/orientation as compared to the case illustrated in FIG. 1 (bent double-pointed arrow in FIG. 2 illustrating a pivoting movement of the leg part **110**). It is understood that both leg parts **110** are pivotally connected to the rest of the bipod **1** in a corresponding manner.

(49) According to said quick-release interface connection aspect of the present invention, said interface connection adapter arrangement **300** comprises a first part **310** and a second part **320**, the first part **310** and the second part **320** being translatable in relation to each other along a movement dimension **380** (see FIG. 12), the movement dimension **380** possibly being in or parallel to said main connection plane **303**.

(50) Further according to said quick-release interface connection aspect, the interface connection adapter arrangement **300** comprises a first rifle fastener **331**, in turn comprising cooperating fastener means both on said first part **310** and on said second part **320**. The first rifle fastener **331** is arranged to be activated, by moving the first part **310** in relation to the second part **320** along the movement dimension **380**, for fastening the interface connection adapter arrangement **300** to the rifle **2**. In particular, the activation takes part by translating the first part **310** in relation to the

second part **320** into a gripping position, in which the first rifle fastener **331** releasably engages with the rifle **2** connection interface as mentioned above by gripping said rifle **2** connection interface.

(51) As illustrated in the Figures, the interface connection adapter arrangement **300** is arranged to be connected to said two leg parts **110**.

(52) First Aspect of the Invention: A Detachable Leg Arrangement

(53) Turning now specifically to said detachable leg arrangement aspect of the present invention, according to this aspect the leg part attachment means **120** comprises a mushroom-shaped support part **123** with an abutment surface **124** arranged to face said proximal end **113** of the leg part **110** in question when the leg part attachment means **120** is in said engaged state. See, in particular, FIGS. **4-8** showing a detachable leg arrangement **100** in turn comprising the leg part attachment means **120** as well as the leg part **110** itself and said leg fastening part **130**.

(54) That the leg part attachment means **120** is “mushroom-shaped” means that it comprises a relatively narrow stem part and a relatively wide top part. The abutment surface **124** is then arranged on a side of the top part facing the stem part. Such a mushroom-shape is exemplified in FIG. **8**.

(55) Furthermore according to this aspect, the support part **123** has a shape **125**, such as in a cross-section perpendicularly to said axial direction **111** of the leg part **110** in question, when the leg part attachment means **120** is in said engaged state with the leg part **110**, the shape **125** being a non-circular shape. In particular, it is a top part of said mushroom-shape of the support part **123** that has said shape **125**. The shape **125** may be a cam shape, arranged to interact with a particular corresponding limiting wall shape of a space **131** of said leg fastening part **130** arranged to receive the support part **123**. The shape **125** may be defined in a plane being perpendicular, or substantially perpendicular, to the axial direction **111** of the leg part **110** in question, and/or it may be defined in a plane being a turning plane used for engaging and disengaging the support part **123** in relation to the leg fastening part **130** by screwing the leg part **110** as described below. It is preferred that the leg part **110** is turned about its axis **111** when engaging and disengaging the support part **123** in relation to the leg fastening part **130**.

(56) Since the shape **125** is non-circular, it can engage with a corresponding shape of said leg fastening part **130**, such as engaging with a shape of said space **131**, so as to limit a rotational movement of freedom of said leg part attachment means **120** in relation to the leg fastening part **130** when the leg part attachment means **120** is received into and accommodated in the space **131**, this rotational movement of freedom being in said shape-defining plane. Hence, when the leg part attachment means **120** is in said engaged state and the leg part attachment means **120** also engages the leg fastening part **130**, as is illustrated in FIG. **5**, a rotation of the leg part **110** in relation to the leg fastening part **130** will cause the threaded engagement between the leg part **110** and the leg part attachment means **120** to deepen or loosen, by the respective threads interacting, thereby bringing the leg part attachment means **120** axially **111** towards or away from the leg part **110**.

(57) By bringing the leg part attachment means **120** axially towards the leg part **110** by screwing the leg part **110**, the abutment surface **124** of the leg part attachment means **120** moves towards the leg part **110**, and can be pushed axially onto a corresponding abutment surface **132** (axially **111** facing away from the leg part **110**) of the leg fastening part **130**.

(58) Thereby, the leg fastening part **130**, such as a flange of the fastening part **130**, may be pressed between the proximal end **113** of the leg part **110** and said abutment surface **124** of the leg part attachment means **120** and as a result fasten the leg part **110** to the leg fastening part **130** using a pressing/friction-based engagement of the fastening part **130** between the leg part **110** and the leg part attachment means **120**. By again unscrewing the leg part **110** out of a completely fastened threaded engagement with the leg part attachment means **120**, which is again performed by rotating the leg part **110** in relation to the leg fastening part **130** but in an opposite thread loosening direction, the leg part attachment means **120** again moves axially **111** away from the leg part **110**

and as a result the abutment surface **124** releases said pressing engagement with the leg fastening part **130**.

(59) It is noted that said non-circular shape **125** then admits both engaging and disengaging of said pressing engagement connecting the leg part **110** to the leg fastening part **130** by simply rotating or screwing the leg part **110** in relation to the leg fastening part **130**. In the preferred case in which the leg fastening part **130** is attached or attachable to the rest of the bipod, this means that the leg part **110** can be rotated or screwed in relation to the rest of the bipod **1** to fasten and unfasten the leg part **110** conveniently.

(60) As is perhaps best illustrated in FIG. **8**, said shape **125** comprises two parallel, opposed side edge parts (sections) **126**, that may further be parallel to a leg part attachment means **120** insertion direction **135**, in which the leg part attachment means **120** is inserted into said space **131** when the leg part **110** is to be attached to the leg fastening part **130**.

(61) Furthermore, the shape **125** may comprise an edge part (section) having a particular non-circular cam shape, such as a curved shape, such as an arc-shaped (as illustrated in FIG. **8**) edge part **127**, connecting said two side edge parts **126**. The cam shape edge part **127** may, alternatively, have a polygonal or any other suitable shape, even if the present inventor has discovered that a shape achieving a smooth connection between the two side edge parts **126**, without any sharp corners, is useful for quick aligning of the support part **123** with the leg fastening part **130** when the mushroom-head of the support part **123** is inserted into the space **131**.

(62) In particular, the cam shape **125** is designed to be complementary to a corresponding shape at a bottom end of the space **131** of the leg fastening part **130**. As mentioned, the space **131** is arranged to receive and accommodate the mushroom-shaped support part **123**, and specifically said mushroom head as illustrated in FIGS. **5** and **6**.

(63) Furthermore, the shape **125** may comprise a straight edge part (section) **128** connecting said two side edge parts **126**. The straight edge part **128** may then be opposite to the cam shape edge part **127** with respect to said parallel edge parts **126**. The straight edge part **128** may be arranged to, when the support part **126** is completely inserted into said space **131**, align with an exterior shape of the leg fastening part **130** so as to provide a quick way for a user of the bipod **1** to verify that such complete insertion has been accomplished.

(64) As mentioned above, the leg part **110** in question may comprise said first threaded engagement means **114**, arranged to engage with the corresponding second threaded engagement means **121** of the leg part attachment means **120**. In some embodiments, the first engagement means **114** then comprises an axial hole **115** at the proximal end **113** of the leg part **110**, the axial hole **115** being arranged with internal threads arranged to threadedly engage with corresponding external threads of the second engagement means **121**.

(65) Correspondingly then, said second engagement means **121** may comprise an axial bolt **122** with corresponding external threads, so that the mushroom-shaped support part **123** can be axially **111** screwed into the proximal end **113** of the leg part **110** in question. The axial bolt **122** may be a part of the mushroom-shaped support part **123**, and in particular form a stem of said mushroom shape.

(66) In order to further increase the friction provided by said pressing engagement between the leg part **110** and the leg fastening part **130**, the proximal end **113** of the leg part **110** may comprise a washer **116**, forming a proximal end surface of the leg part **110**. The washer **116** may be rotatably engaging with the leg part **110**. The washer **116** may be made from plastic material, such as a thermoplastic material. The washer **116** may form a proximal-end **113** abutment surface arranged to abut, and press, directly against a side of the leg fastening part **130** facing the leg part **110** when said pressing engagement between the leg part **110** and the leg fastening part **130** is in place, hence creating an increasing friction between the leg part **110** and the leg fastening part **130** as the threaded pressing engagement increases by screwing the leg part **110** in relation to the leg fastening part **130**.

(67) FIGS. **18-20** illustrate a leg part locking mechanism according to one embodiment of the present invention. The leg part locking mechanism is active to prevent the leg part **110** from coming loose from the leg part attachment means **120** once fastened, unless the leg part locking mechanism is activated to allow such coming loose of the leg part **110** in relation to the leg part attachment means **120**.

(68) More particularly, according to this embodiment the detachable leg arrangement **100** further comprises an angularly stopper **117**, in turn preferably being angularly movable and comprising teeth **117b**. The angular stopper **117** may be a part of the leg part **110**, as is the case with the teeth **117b**.

(69) The detachable leg arrangement **100** may further comprise an axial stopper **129**, in turn preferably being axially movable and comprising an axial pin **129a**. The stopper **129** and the axial pin **129a** may then form part of the leg part attachment means **120**.

(70) It is realized that the angular stopper **117** could also form part of the leg part attachment means **120**, and the stopper **129** may form part of the leg part **110**.

(71) At any rate, the angular stopper **117**, and specifically said teeth **117b**, is angularly movable in relation to the stopper **129**. The stopper **129**, and specifically the axial pin **129a**, is axially movable in relation to the angular stopper **117**, “angularly” and “axially” being defined in relation to the axial direction **111** of the leg part **110**.

(72) Furthermore, the angular stopper **117**, in particular said teeth **117b**, and the stopper **129**, in particular axial pin **129a**, are mutually axially spring-loaded, such as using a spiral or other type of spring **117a**, in relation to each other, pressing the axial pin **129a** and said teeth **117b** into engagement with each other. This engagement between the axial pin **129a** and the teeth **117b** results in that the leg part **110** is prevented from angularly pivot in relation to the leg part attachment means **120** out of engagement between the leg part **110** and the leg part attachment means **120**.

(73) It is noted that FIG. **19** discloses the locking pin **117c'''** as engaging with the spring **117a** at a point along the length of the spring **117a** not being an end of the spring **117a**. However, it is envisioned that the locking pin **117c'''** may alternatively engage with the spring **117a** in other ways, such as the locking pin **117c'''** engaging with a proximal (in relation to the leg part **110** in question) longitudinal end of the spring **117a**. What is important is that the locking pin **117c'''** is pressed by the spring **117a** in the longitudinal direction **301** away from the distal end **112** of the leg part **110** in question, from a bottom of a hole in the leg part **110** in which the spring **117a** is arranged and which supports the spring **117a** longitudinally, limiting a longitudinal freedom of motion of the spring **117a** towards said distal end **112**.

(74) In other words, once the leg part **110** has been fastened, such as by screwing the leg part **110** thereby bringing the leg part attachment means **120** axially towards the leg part **110** as described above, the engagement between axial pin **129a** and the teeth **117b** prevents the leg part **110** to come loose or undone in relation to the leg part attachment means **120**. Preferably, said spring-loading forces the axial pin **129a** into said engagement with the teeth **117b** as a result of the leg part **110** being moved into a fastening engagement with the leg part attachment means **120**, so that said engagement between the axial pin **129a** and the teeth **117b** is maintained, due to said spring-loading, as long as the leg part **110** remains fastened to the leg part attachment means **120**.

(75) As illustrated in FIGS. **18-20**, the stopper **129** comprises the axial pin **129a**, in turn having an end part **129b** having a shape specifically arranged to engage with the teeth **117b**, and preferably so that the axial pin **129a** can move angularly in relation to the teeth **117b** in a first angular direction but not in a second angular direction. The first angular direction is a direction of the leg part **110** in relation to the leg part attachment means **120** along which the leg part **110** is moved into a deeper fastening engagement with the leg part attachment means **120**, the engagement being as described above. The second angular direction is an opposite angular direction, along which the leg part **110** is moved out of said fastening engagement.

(76) The axial pin **129a** may be axially adjustable, irrespectively of said spring **117a**. For instance, as illustrated in FIGS. **18-30**, the stopper **129** may further comprise a locking screw **129c**, arranged to act on the axial pin **129a**, for instance so as to achieve lock an axial position of the axial pin **129a** in relation to the leg part attachment means **120**. For instance, the locking screw **129c** may be terminate, and be reachable for adjustment, from said space **131**. The locking screw **129a** may be substantially or completely perpendicular to the axial direction **111**.

(77) The axial pin **129a** and/or the locking screw **129c** may run in holes or channels, such as drilled holes, in the leg part attachment means **120**. The axial pin **129a** may have a non-circular cross-section, arranged to interact with a corresponding cross-sectional shape of said hole or channel so as to prevent the axial pin **129a** to pivot about its longitudinal axis. Alternatively or supplementally, the locking screw **129c** may be arranged to lock a pivotal orientation of the axial pin **129a** in relation to its hole or channel, such as by being screw-fastened against a flat surface of the axial pin **129a**.

(78) As is best illustrated in FIG. **20**, each of the teeth **117b** comprises a respective sliding surface **117b'** and a respective stopping surface **117b''**. The sliding surface **117b'** may have an angle of more than 45° in relation to the axial direction **111**, while the sliding stopping surface **117b''** may be parallel or substantially parallel to the axial direction **111**. The end part **129b** may be arranged with a corresponding shape so as to achieve the above-described engagement behaving different in said different angular directions.

(79) Moreover, the detachable leg arrangement **100** may further comprise an axial actuator **117c**, arranged to, when axially activated, pull said axial pin **129a** and said teeth **117** apart, against said spring-loading force, the spring-loading force possibly being due to the spring **117a** as described above.

(80) In FIGS. **18-20**, this is exemplified by an inner collar **117c'** engaging with the spring **117a** via locking pin **117c'''** running through a through hole **117c''''** of the leg part **110**. The inner collar **117c'** comprises the teeth **117b**, and is slidably provided on and along the leg part **110**. The through hole **117c''''** allows a certain axial freedom of movement of the inner collar **117c'** in relation to the rest of the leg part **110**.

(81) An outer collar **117c''** is fastened to the inner collar **117c'**, hiding the locking pin **117c'''** and being shaped with a friction-increasing grip part allowing a user to pull the outer collar **117c''** and hence the teeth **117b**, in relation to the rest of the leg part **110**, axially away from the leg part attachment means **120**, and hence also away from the stopper **129**. This way, by pulling the outer collar **117c''** against the spring force of spring **117a**, the user can temporarily disengage the teeth **117b** from the axial pin **129a**, allowing the fastened leg part **110** from being unfastened from the leg part attachment means **120** by unscrewing the leg part **110** as described above.

(82) Hence, when attaching the leg part **110** to the rest of the leg arrangement **100**, the leg part **110** is screwed into engagement with the leg part attachment means **120**, in the general way described above in relation to FIGS. **1-13**. As this engagement tightens, the spring **117a** further presses the teeth **117b** into engagement with the axial pin **129a**. The screwing of the leg part **110** can continue until full engagement between the leg part **110** and the leg part attachment means **120**, due to an axial freedom of movement allowed by the spring **117a** and due to the properties of the teeth-pin engagement as described above. Once the leg part **110** is fully engaged with the leg part attachment means **120**, it can only be loosened by pulling the teeth **117b** axially apart from the axial pin **129a**, by axially pulling the outer collar **117c''** in the described way. Once the collar **117c''** has been axially pulled, the leg part **110** can be unscrewed from the leg part attachment means **120**.

(83) The axial pin **129a** can be mounted as a part of the leg part attachment means **120** irrespectively of if the leg part **110** comprises said angular stopper **117** or not, providing for a more flexibly usable leg part attachment means **120**.

(84) As mentioned, the detachable leg arrangement **100** may comprise the leg fastening part **130**, in turn comprising the space **131** arranged to receive and accommodate the mushroom-shaped support

part **123**. As also discussed, the space **131** may have the abutment surface **132**, in turn arranged to engage with the abutment surface **124** of the mushroom-shaped support part **123** so that said abutment surfaces **124**, **132** engage with a mutual friction engagement as the leg part attachment means **120** is pressed axially **111** towards the leg part **110** when the leg part attachment means **120** is in said engaged state and the leg part **110** is screwed in relation to the leg fastening part **130** thereby deepening said pressing engagement. Then, the leg part **110** is fixed in relation to the leg fastening part **130** until the friction engagement is loosened by turning the leg part **110** in the opposite direction.

(85) Moreover, the space **131** may have an end opening **133** arranged to receive the leg part attachment means **120** into the space **131**, when the leg part attachment means **120** is in said engaged state. Then, the leg part attachment means **120** is received into said space **131** in the above-mentioned insertion direction **135**, in turn being perpendicular to said axial direction **111** of the leg part **110** in question.

(86) The space **131** may furthermore have an elongated opening **134**, extending across the abutment surface **132** of the leg fastening part **130** from said end opening **134** and preferably along said insertion direction **135**. Then, the elongated opening **134** may be arranged to accommodate an axial part of the leg part **110** and/or of the leg part attachment means **120**, such as the axial bolt **122** as is best illustrated in FIG. 6.

(87) The elongated opening **134** may be arranged with a bottom end **136**, arranged to limit the movement, in said insertion direction **135**, of the axial bolt **122** or support part **123**, in relation to the leg fastening part **130**. When inserting the correctly-oriented support part **123** into the space **131** in the insertion direction **135**, the leg fastening part **130** may be arranged so that the axial bolt **122** reaches the bottom end **136** of the elongated opening **134** at the same time as the cam shape part **127** reaches the bottom end of the space **131**. This provides a convenient way of avoiding misalignment of the inserted support part **123** with the leg fastening part **130**.

(88) The leg part attachment means may be made from metal material, such as steel or aluminium, and may be constituted by a single, integrated piece of metal material.

(89) As discussed above, the leg fastening part **130** in turn may further comprise a fastening means, such as an adjustable leg fastening arrangement **200** of the present type, arranged to releasably and/or movably fasten the leg fastening part **100** to a bipod main part, such as an interface connection adapter arrangement **300** of the present type. In other embodiments, the fastening part **130** may constitute an integrated part of the rest of the bipod **1**.

(90) As described above, the present solution according to said detachable leg attachment aspect is useful for quickly attaching and detaching a leg part **110** to the rest of the bipod **1**. In some embodiments, the invention also relates to a leg kit for the bipod **1**, the kit comprising a detachable leg arrangement **100** of the type described herein, in turn comprising one or a pair of leg parts **110** of the present type. Such a kit may then further comprise one or several additional leg parts **110**, or one or several additional pairs of additional leg parts **110**. Then, said additional leg parts **110**, or pairs, have different properties in terms of length, length adjustability, flex, material, weight and/or foot parts **117**, as compared to the one or two leg parts **110** already forming part of the detachable leg arrangement **100**. All leg parts **110** included in such a kit are then compatible with the leg fastening part that also includes the leg fastening part **130** of the bipod **1**, for fast and convenient replacement of the leg parts **110** in question depending on need. For instance, each such leg part **110** may comprise its own respective leg part attachment means **120**, which may be in threaded engagement with the leg part **110** in question as described above and ready to be attached to the leg fastening part **130** by inserting and screwing as described above.

(91) Second Aspect of the Invention: An Adjustable Leg Fastening Arrangement

(92) Turning now explicitly to the adjustable leg fastening aspect of the present invention, and in particular to FIGS. 9-11, said adjustable leg fastening arrangement **200** may comprise a conical recess **211** and a conical part **221**, an outer surface of the conical part **221** being arranged to engage

in a friction engagement with an inner surface of said conical recess **211**. Hence, the conical shape of both recess **211** and part **221** correspond to each other so that such friction engagement is possible between the two. For instance, it is preferred that the recess **211** and part **221** both have the same or at least substantially the same cone angle. Furthermore, the cone radii of the recess **211** and part **221** are adapted so that the conical part **221** fits at least partly (such as completely) within the conical recess **211** in a way allowing said outer surface to be in direct contact with said inner surface. Preferably, such a direct contact surface is in total at least 1 cm.^{sup.2} when the cone part **221** is inserted into the conical recess **211** as far as possible.

(93) Said cone angle may, for instance, be between 20° and 40°, such as between 25° and 35°, in relation to a cone axis of the conical part **221** (its centrum axis).

(94) Moreover, the conical recess **211** and the conical part **221** are each arranged to be rotationally-fixedly attached to a respective different one of said first pivoting part **210** and said second pivoting part **220**.

(95) Herein, by two parts being arranged “rotationally-fixedly” in relation to each other is meant that the two parts cannot rotate in relation to each other, in particular not about a common rotation axis. For the conical recess **211** and the conical part **221** in particular, this means that neither of these two parts **211**, **221** can rotate, about a respective cone axis of the respective cone shape, in relation to the first/second pivoting part **210/220** to which it is fixed or of which it forms a part.

(96) The first pivoting part **210** can rotate in relation to the second pivoting part **220**, and in particular this rotational freedom of movement exists with respect to a main axis of the respective cone shape of the conical recess **211** and the conical part **221** when the part **221** is aligned with, and inserted into, the recess **211** and the adjustable leg fastening arrangement **200** as a result is in an assembled state. The cone axis is then the same for the respective cone shape of the conical recess **211** and the conical part **221**, as a consequence of the automatic alignment achieved by pressing the conical part **221** into the conical recess **211** until abutment between said two conical surfaces and/or due to other alignment means arranged on the first pivoting part **210** and/or on the second pivoting part **220** (such as the below-described pin **223** cooperating with the below-described through holes **222**). As a result, the conical recess **211** and the conical part **221** pivot in relation to each other, about said common cone axis, as said first pivoting part **210** pivots in relation to said second pivoting part **220** and when the adjustable leg fastening arrangement **200** is in said assembled state. In case the conical part **221** is pressed against the conical recess **211**, such pivoting will result in friction between said abutting surfaces, the friction increasing as a function of a force pressing the conical part **221** cone-axially towards the conical recess **211**.

(97) Then, the adjustable leg fastening arrangement **200** further comprises a screw engagement means **201** arranged to, when deepening a screw (thread) engagement of said screw engagement means **201**, press the conical part **221** axially (“axially” here referring to said cone axis) into and against said inner surface of the conical recess **211**, thereby increasing a friction of said friction engagement between the inner surface of the conical recess **211** and the outer surface of said conical part **221**.

(98) This way, by tightening the screw engagement means **201** to a selected tightening setting, such as to a selected angular distance and/or to a selected torque, a desired corresponding surface friction will result with respect to said friction engagement, in turn deciding a corresponding force required to turn the leg part **110** being fastened to the rest of the bipod **1** using the adjustable leg fastening arrangement **200**.

(99) Hence, a user operating said screw engagement means **201** can easily select, for instance, a desired friction to be able to adjust a pivot angle of the leg part **110** by hand so as to adjust a height or inclination of the attached rifle **2** in the way discussed above, without the rifle **2** coming down due to excess pivoting of the leg part in question **110**. Once content, the user can tighten the screw engagement means **201** further, to increase the friction and as a result lock the leg part **110** in question in the selected pivot orientation in order to use the rifle **2** without the bipod **1** yielding. As

will be described below, in connection to FIG. 15, such “locking” may simply imply a sufficiently increased friction engagement.

(100) For transport and storage, the screw engagement means **201** may again be loosened so as to allow the leg part **110** to easily be pivoted to a desired folded orientation, and if so desired the leg part **110** can be locked such a folded orientation by again tightening the screw engagement means **201**.

(101) The corresponding is true for both leg parts **110** of the bipod **1**, since the bipod **1** comprises one adjustable leg fastening arrangement **200** of the type discussed herein for each leg part **110**. A respective friction engagement of each of the adjustable leg fastening arrangements **200** can preferably be individually adjustable using a corresponding screw engagement means **201**, allowing the user to individually set the friction engagement properties for each leg part **110**. For instance, by loosening the friction engagement of one leg part **110**, that leg part **110** can be adjusted while not affecting the current pivot angle of the other leg part **110**.

(102) It is noted that the friction engagement is not only capable of being set to a “pivotable”/“non-pivotable” state, but (due to the threaded engagement of the screw engagement means **201**) can instead be set across a continuous interval of possible friction settings. This is very useful, since it allows the user to set a desired fine-tuned friction of the friction engagement, and hence a required torque to pivot the leg part **110** in question, very accurately and depending on circumstance. For instance, fine-tuning of the rifle **2** position may require a higher friction as compare do coarse pivoting; different rifles **2** may weight differently, requiring different leg part **110** pivot frictions for manipulation; leg parts **110** of different lengths may require different pivot frictions depending on where along the leg part **110** in question the user applies a pivoting force; and so forth.

(103) The continuously settable friction of the continuous pivoting freedom of motion of each leg part **110** admits full adjustability of the bipod **1** even in case the leg parts **110** themselves are not adjustable with respect to leg part **110** axial **111** length (which is preferred). Hence, each leg part **110** can be made very rigid and stable while still achieving full bipod **1** adjustability. Each individual leg part **110** may, for instance, be formed as one integrated piece of metal material, such as steel or aluminium, providing a very sturdy bipod **1**.

(104) In some embodiments, such as the exemplifying one shown in the Figures, said conical recess **211** is a part of said first pivoting part **210**, such as the conical recess **211** being an integrated part of an integrated material body forming the whole or a part of the first pivoting part **210**. Moreover, said conical part **221** may be a part of the second pivoting part **220**, such as a separate part fixed to the second pivoting part **220** using said screw engagement means **201**.

(105) In some embodiments, the first pivoting part **120** comprises a single integrated piece of metal material comprising both said conical recess **211** and said leg fastening part **130**, in turn being arranged to fasten said leg part **110**, for instance in the above-described manner.

(106) The screw engagement means **201** may be a part of the second pivoting part **220**. It may further comprise an axial hole **224** with internal threads, arranged to threadedly engage with external threads of a pressing screw **226** which is also comprised in the screw engagement means **201**. By tightening this screw engagement, the friction between said conical surfaces may then increase, as described above. The pressing screw **226** may be arranged with a gripping head or, as illustrated in the Figures, a tool-engagement head such as an Allen key hole.

(107) Furthermore, the pressing screw **226** may be arranged to run along an axial through hole **222** through the conical part **221**. Then, the axial hole **224** may for instance be arranged at a narrow-end cone-axial side of the conical part **221**, while the pressing screw **226** runs through said axial through hole **222** to an opposite, wide-end cone-axial side of the conical part **221** at which the pressing screw **226** by turning applies an axial pressing force onto the conical part **221** against the conical recess **211**.

(108) Then, the axial through hole **222** through the conical part **221** may have a non-circular cross-section, such as the hexagonal cross-section used in the exemplifying embodiment shown in the

Figures (see, in particular, FIG. 4). The second pivoting part **220**, in turn, then may further comprise a pin **223** arranged to slidably engage with (inside) said axial through hole **222** and engage with said non-circular cross-section so as to limit a rotational movement of the conical part **221** in relation to the pin **223** about the cone axis. For instance, the pin **223** may have a cross-section corresponding to, or at least being arranged to engage in a way allowing no rotation, about the above-discussed conical shape cone axis, of the conical part **221** in relation to the pin **223**. The Figures (again, see FIG. 4) illustrates the example in which the pin **223** has a corresponding hexagonal outer cross-sectional periphery as the hexagonal interior cross-sectional periphery of the axial through hole **222** through the conical part **221**.

(109) In particular, and is illustrated in the Figures, the pin **223** may comprise the axial hole **224** with internal threads.

(110) Hence, the second pivoting part **220** may comprise both the pin, the conical part **221** and the pressing screw **226**, whereby the pressing screw is arranged to be inserted axially through the conical part **221**, via through hole **222**, and fasten by threading the screw into the axial hole **224** of the pin **223**. This way, the conical part **221** is slid onto the pin **223**, the described cross-sectional shapes of the conical part **221** through hole **222** and the pin **223** engaging so as to prevent any rotation of the conical part **221** in relation to the pin **223** about the cone axis, as described. The pressing screw **226** holds the conical part **221** cone-axially in place and determines the pressing force in turn giving rise to the selected leg part **110** pivoting friction discussed above.

(111) This achieves that the conical part **221** is rotationally-fixed in relation to the second pivoting part **220**, along with the pressing screw **226** (apart from the pressing screw **226** being screwable to increase or decrease the selected friction). Hence, when pivoting the leg part **110** in question back and forth, the screw engagement defining the applied pivoting friction is not affected. In other words, once the user has selected a particular friction by adjusting the pressing screw **226**, the friction remains the same until the pressing screw **226** is again adjusted, even under vigorous pivoting of the leg part **110** in question.

(112) It is noted that the conical part **221** is hence rotationally-fixed in relation to the second pivoting part **220**. However, in the possible case in which the conical part **221** is a part of the first pivoting part **210** (and the conical recess **211** instead being part of the second pivoting part **220**), the screw engagement means **201** may correspondingly instead be effective to rotationally-fixed attach the conical part **221** to the first pivoting part **210**.

(113) It is further realised that other mechanisms than the one illustrated in the Figures can be implemented to provide said rotationally-fixed engagement between the conical part **221** and the pivoting part **210** or **220** in question, such as the conical part **221** having a protruding part at its end having an outer non-circular cross-section arranged to engage to provide cone-axial rotational fixing in relation to a corresponding hole with a corresponding interior cross-section of the pivoting part **210** or **220** in question. In other examples, the conical part **221** and/or the pivoting part **210** or **220** in question comprises an eccentrically-located (in relation to said cone axis) stud arranged to engage with a corresponding eccentric hole to achieve said rotational fixing. What is important is that the screw engagement means **201** comprises a fixing means arranged to allow the conical part **221** to move along the cone axis as the screw engagement means **201** is adjusted to achieve a selected pivoting friction, but so that the conical part **221** is at the same time rotationally-fixed in relation to the pivoting part **210** or **220** not comprising the conical recess **211**.

(114) In some embodiments, the second pivoting part **220** further comprises a pressing washer **225**. The pressing screw **226** is then arranged to press the pressing washer **225** cone-axially onto the conical part **221**, thereby pressing the conical part **221** cone-axially towards, into and against said conical recess **211**. The pressing washer **225** preferably has a larger direct or indirect contact surface against the conical part **221** than a cone-axial cross-section of the pressing screw **226**, which provides for a better force distribution when adjusting the pressing screw **226** to achieve a selected friction. The pressing screw **226** may run through a cone-axial through hole of the pressing

washer **225**.

(115) As is illustrated in the Figure, the pressing washer **225** may have a generally tapered shape, with a relatively narrow part facing away from the conical part **221** and a relatively wide part facing towards the conical part **221**. This provides a balanced force distribution. The pressing washer **225** may be circular-symmetric about said cone axis.

(116) The pressing washer **225** may be made of plastic or, preferably, metal material. In some embodiments, the pressing washer **225** is made from an integrated piece of aluminium bronze material.

(117) Furthermore, the conical recess **211** may have a conical inner metal surface arranged to engage with said conical part **221**, in particular in case the conical recess **211** is formed as a conical recess in the very metal material constituting the integrated metal body forming the pivoting part **210** or **220** in question. However, in this and in other embodiments the conical part **221** may have a conical outer plastic surface being arranged to engage with said conical recess **211**. The conical part **221** may even be made in its entirety from a plastic material, such as an integrated piece of plastic material. Suitable plastic materials comprise thermoplastic materials, such as elastic thermoplastic materials, for instance POM (PolyOxiMetylen), sold with different properties under trade names Acetal, Polyacetal, Delrin, Hostaform, Kepital and Tenac.

(118) With respect to said pivoting movement of the leg part **110** in relation to the rest of the bipod **1**, the first pivoting part **210** may comprise or be connected to a leg fastening part **130** of the above-discussed or other types, in turn being arranged to fasten the leg part **110** so that its axial direction **111** is neither perpendicular nor parallel to an axial direction of said friction engagement (the cone-axial direction about which the leg part **110** will pivot). In other words, the pivoting movement of the leg part **110** in relation to the second pivoting part **220** describes itself a cone-shape if pivoted over a full 360°, a cone axis of the described cone being identical to the cone axis of the conical part **221** and the conical recess **211**. Such described cone-shape will advantageously have an angle, in relation to said cone axis, of between 60° and 80°, such as between 65° and 75°.

(119) As described above, the leg fastening part **130** may be arranged to detachably fasten the leg part **110** in a fixed position relative to said leg fastening part **130**, such as once screwed into place and hence held by said friction engagement between abutment surfaces **124**, **132**.

(120) In some embodiments, the second pivoting part **220** comprises fastening means **227**, arranged to detachably and fixedly fasten the second pivoting part **220** to the rest of the bipod **1**, such as to the bipod main part, or directly to the rifle **2**. As can be seen, for instance, in FIG. **10**, the fastening means **227** may comprise at least two through holes **228** and corresponding fastening screws **229**.

(121) Note that FIGS. **10a** and **10b** show exploded perspective views of the same parts, but whereas FIG. **10a** shows the parts in a more assembled state, FIG. **10b** provides a more accurate explosion of the subparts. In particular, FIG. **10b** shows the mechanical axial relation between the pressing washer **225**, the conical part **221**, the first pivoting part **220** and the bearing **230** more correctly than FIG. **10a**.

(122) The first pivoting part **210** may furthermore be pivotally connected to the second pivoting part **220**, not only via the aggregate formed by the conical part **221** and the conical recess **211**, but also via a friction-reducing bearing **230**, such as a circular-symmetric piece of plastic material, such as a thermoplastic material, for instance polyoxymethylene. The bearing **230**, which may be slightly resilient in the cone-axial direction, may then be pressed between the pivoting parts **210**, **220** as the conical part **221** is pressed towards the conical recess **211**.

(123) As is perhaps best illustrated in FIG. **12**, said fastening means **227** may be arranged to fasten the second pivoting part **220** to the bipod main part, or the rifle **2**, in at least a first attachment orientation and a second, different, attachment orientation in relation to the attached rifle **2**. More particularly, a cone axis of said pivot cone described by the leg part **110** when pivoting, in relation to the conical recess **211** and hence in relation to the attached rifle **2**, may have a different angle as compared between said first and second different attachment orientations. In particular, an angle of

said described pivot cone may vary, across said two different attachment orientations, completely or substantially in a plane parallel to a main longitudinal or shooting direction of the attached rifle (124) Said two different attachment orientations are illustrated, by way of example, in FIG. 12, where the interface connection adapter arrangement **300** comprises three through holes **312** for each of the two adjustable leg fastening arrangements **200**, arranged so that the two through holes **228** can be connected to either of a front or a rear hole pair formed by the three holes **312** in question, using the fastening screws **229**. When the first pair of holes **312** is used (first attachment orientation), the leg part **110** pivot cone will have a cone axis the angle of which, in a plane spanned by the three holes **312**, is different from a cone axis of the cone described by the pivoting leg part **110** as attached using the other pair of holes **312** (second attachment orientation).

(125) In particular, said first attachment orientation may allow the leg part **110** to be pivoted to an orientation in which the leg part **110** axis is completely or substantially parallel to a longitudinal or shooting direction or axis of the rifle **2**, while the second orientation does not allow the same.

(126) The hole pair used in said second attachment orientation may instead be arranged along a line which is parallel to a corresponding line on the other side of the bipod **1**, arranged to fasten the other leg part **110** thereto. This allows the angle between the leg parts **110** to be constant in any corresponding pivot orientation of both the leg parts **110** in question, at least in the preferred case in which a pivot axis of both leg parts **110** are parallel and preferably identical when using said second attachment orientation.

(127) Third Aspect of the Invention: A Reversible Interface Adapter Arrangement

(128) Turning now explicitly to the reversible interface connection aspect of the present invention, reference is particularly made to FIG. 12.

(129) In this aspect, the above-mentioned interface connection adapter arrangement **300** further comprises said first interface side **330**, arranged on said first side of the main connection plane **303**, and said second interface side **340**, arranged on said second side of the main connection plane **303**.

(130) The first interface side **330** is then provided with a first rifle fastener **331**, the first rifle fastener **331** in turn being arranged to fasten the interface connection adapter arrangement **300** to the rifle **2** in accordance with a first rifle fastening system.

(131) Moreover according to this aspect, the second interface side **340** is provided with a second rifle fastener **341**, the second rifle fastener **341** being arranged to fasten the interface connection adapter arrangement **300** to the rifle **2** in accordance with a second rifle fastening system.

(132) Then, said second rifle fastening system is associated with a different fastener geometry as compared to said first rifle connection system.

(133) In combination with the above-described leg parts **110** being able to selectively pivot to extend on either side of the main connection plane **303**, such an interface connection adapter arrangement **300** achieves a simple but useful way of providing added flexibility to the bipod **1**. By pivoting the leg parts **110** so that they both are directed out from the connection adapter arrangement **300** from said first interface side **330**, the rifle fastening system of the second interface side **340** can be used to fasten the bipod to a particular rifle **2**. When then using the bipod with a different type of rifle fastening system, the leg parts **110** can simply instead be swung to be directed out from the connection adapter arrangement **300** from the second interface side **340**, and then the same bipod **1** can be used with the rifle fastening system of the first interface side **330**.

(134) In other words, one and the same bipod **1** can be used with two different rifle fastening systems, without a user having to add or replace any component parts of the bipod **1**, and in particular not having to replace or demount the leg parts **110**.

(135) In some embodiments, said first rifle fastening system is an ARCA system or a Picatinny system. Both of these systems are per se conventional and well-known rail-type rifle mounting systems. Whereas ARCA allows fastening of an accessory at any position along the rail, Picatinny is typically associated with a plurality predetermined discrete fastening positions. It is noted that other rifle fastening systems are also known.

(136) Similarly, the second rifle fastening system may also be an ARCA system or a Picatinny system, however the first and second rifle fastening systems are always different in terms of their fastening geometry.

(137) That the fastening systems are different in terms of their “fastening geometry” means that the first rifle fastening system and the second rifle fastening system are incompatible, in the sense that the first rifle fastener **331** cannot be used to fasten the bipod **1** to the rifle **2** using the second rifle fastening system and/or that the second rifle fastener **341** cannot be used to fasten the bipod **1** to the rifle **2** using the first rifle fastening system.

(138) As mentioned above, the interface connection adapter arrangement **300** may comprise the first part **310** and the second part **320**, being translatable in relation to each other along the movement dimension **380** in the main connection plane **303**.

(139) Furthermore as discussed above, the interface connection adapter arrangement **300** may comprise the first rifle fastener **331**, in turn comprising cooperating fastener means both on said first part **310** and on said second part **320**, and being arranged to be activated, by moving/translating the first part **310** in relation to the second part **320** along the movement dimension **380**, such as into a gripping position as described above, for fastening the interface connection adapter arrangement **300** to the rifle **2**.

(140) In a corresponding manner, the second rifle fastener may also comprise cooperating fastener means both on the first part **310** and on the second part **320**. Then, both said first rifle fastener and said second rifle fastener are arranged to be activated for fastening the interface connection adapter arrangement **300** to the rifle by translating said first part **310** in relation to said second part **320** into a respective gripping position.

(141) As is illustrated in FIG. **12**, the first part **310** and the second part **320** are translatable in relation to each other in a direction (along movement dimension **380**) having a non-zero component that is parallel to a line interconnecting the respective pivot points of the two leg parts **110**. Preferably, said direction also has a non-zero component that is perpendicular to said line interconnecting said leg part **110** pivot points, such as such a perpendicular line further being parallel to said main connection plane **303**. In other words, both rifle fasteners may preferably be activated by translating them in relation to each other, into said gripping position, along a curved or (preferably) straight translation line which is obliquely set in relation to a line drawn between the respective pivot points of said leg parts **110**, and furthermore obliquely set in relation to both the longitudinal direction **301** and the lateral direction **302**. Hence, the translation will normally take place along a translation line which is not perpendicular to a shooting direction of the rifle **2**. In all cases, it is preferred that said translation line runs in a plane parallel to said main connection plane **303**.

(142) Using such an oblique translation line conveniently avoids problems with the so-called mechanical “drawer effect”, guaranteeing a smooth operation of the translation without any interlocking between the parts.

(143) The present inventors have discovered that, even if the movement dimension **380** may in some embodiments be parallel to the lateral dimension **302** and/or perpendicular to respective abutment surfaces **313**, **323** of the first **310** and second **320** parts (see below), the dimension of movement **380** may instead advantageously be set at an angle of between 5° and 40°, preferably between 10° and 30°, in relation to said longitudinal direction **301** and/or in relation to said abutment surfaces **313**, **323**.

(144) In some embodiments, such as will be exemplified in closer detail below in relation to the fourth aspect of the present invention, the interface connection adapter arrangement **300** may comprise an adjustment means **360** for adjusting the relative translational position of the first part **310** in relation to the second part **320** along said movement direction **380**. By such adjustment, a gripping force applied by the interface connection adapter arrangement **300** is then also adjusted in relation to the rifle **2**, and in particular in relation to said rifle fastening system, such as a fastening

rail attached to the rifle **2**.

(145) Then, the adjustment means **360** may be arranged to press the first part **130** and the second part **320** together, for instance in the way described above using a screw engagement that can be tightened to achieve said gripping force, thereby achieving a fastening of the interface connection adapter arrangement **300** to the rifle **2**, via gripping engagement of the parts **310**, **320** and the rifle fastening system.

(146) In particular, said adjustment means **360** may be arranged to apply a pressing force of the first part **310** in relation to the second part **320** in a plane being parallel to the main connection plane **303**. The pressing force may, for instance, be applied in the movement direction **380** and/or in the lateral direction **302**.

(147) Fourth Aspect of the Invention: A Quick-Release Interface Adapter Arrangement

(148) Turning now explicitly to said quick-release interface connection aspect of the present invention, still with particular reference to FIG. **12**, the interface connection adapter arrangement **300** according to this aspect further comprises a spring means **350**.

(149) The spring means **350** is arranged to press (such as using the pressing force described above in connection to the third aspect) the first part **310** and the second part **320** together along said movement dimension **380**, and as a result bring, using said force, the first rifle fastener **331** into engagement with the rifle **2** (and correspondingly for the second rifle fastener **341** if used instead of the first rifle fastener **331**).

(150) According to this quick-release interface adapter arrangement, however, the interface connection adapter arrangement **300** further comprises a freedom of movement adjustment means **360**, arranged to adjustably limit a freedom of translational movement of said first part **310** in relation to said second part **320** away from each other along said movement dimension **380**.

(151) It is noted that the freedom of movement adjustment means **360** is the same part as the adjustment means **360** described in connection to the third aspect, above, and that the part **360** in the example illustrated in the Figures conveniently has this double function. However, in other embodiments the interface connection adapter arrangement **300** may instead comprise a distinct adjustment means as well as a separate, distinct freedom of movement adjustment means.

(152) Since the spring means **350** acts to press the parts **310**, **320** together, towards and into said gripping position, unless no other counter-acting forces are applied, the interface connection adapter arrangement **300** mounted on the rifle **2** for engagement of the rifle fastener **331** or **341** in question with said rifle fastening system will remain in said gripping position under the influence of the force applied by the spring means **350**. The parts **310**, **320** may be brought out of the gripping position, to disengage from the rifle **2**, by pulling them apart **310**, **320**, against the spring force of the spring means **350**, but this disengaging movement is limited by said freedom of movement adjustment means **360**.

(153) Preferably, the freedom of movement means **360** may be arranged to selectively adjust the freedom of movement of the first part **310** in relation to the second part **320**, and in particular the relative freedom of motion between parts **310**, **320** in relation to said gripping position as a defined extreme end of a relative translational motion. Furthermore, the freedom of movement limitation means **360** may be arranged to limit such relative movement so that the parts **310**, **320** cannot at all move away from the gripping position, resulting in that the parts **310**, **320** are locked in the gripping position due to the movement limiting effect of the thus set freedom of movement adjustment means **360**.

(154) As is illustrated in FIG. **12**, the spring means **350** may be arranged to apply its spring force to the first part **310** in relation to the second part **320** in a force direction which is parallel, or at least substantially parallel (such as at the most 10° different from) a direction along which the freedom of movement limitation means **360** limits the movement of the first part **310** in relation to the second part **320**.

(155) As is further illustrated in FIG. **12**, the translational movement of the first part **310** in relation

to the second part **320** may be guided, such as be limited to relative movement along said movement dimension **380**, by the spring means **350** and/or the freedom of movement limitation means **360** comprising guide means restricting the relative movement of the first part **310** in relation to the second part **320** to translational movement along a curved or (preferably) straight path as defined by said guide means. In FIG. **12**, these guide means are defined by, firstly, the combination of the limiting screw(s) **351** and the limiting screw channel(s) **354**, and, secondly, the combination of bolt **365** and channel **367**.

(156) Such a construction provides a very easily operated and flexible bipod **1**, which may still be very stable during operation to support a rifle **2**. This will be detailed in the following.

(157) In some embodiments, the spring means **350** may comprise a limiting means, providing an ultimate limit to the freedom of movement of the parts **310**, **320** in relation to each other along the movement dimension **380**, in addition to the freedom of movement limitation provided by the freedom of movement adjustment means **360**. In the example illustrated in FIG. **12**, such limiting means comprises a limiting screw **351**, provided in a limiting screw channel **354**, where an outer thread **352** of the limiting screw **351** is arranged to engage with the first part **310** and a screw head **353** of the limiting screw **351** is arranged to engage with the second part **320**, and/or vice versa. The engagement may then be with a corresponding inner thread **355** of the limiting screw channel **354** in question. This way, the limiting screw **351** may hence limit a freedom of movement of the first part **310** in relation to the second part **320** along the movement dimension **380**.

(158) Moreover, the spring means **350** may further comprise a spring **356** acting to press the first part **310** towards the second part **320** along said movement dimension **380** in the way discussed above.

(159) It is noted that, in the embodiment example illustrated in FIG. **12**, the spring means **350** comprises two different springs **356**, these two springs **356** being arranged to apply said spring force pressing together the parts **310**, **320** along two different but parallel force vectors (that can also be parallel to said movement dimension **380**). Then, each such spring **356** may be associated with its own respective limiting screw **351** and limiting screw channel **354**. The springs **356**, the screws **351** and/or the screw channels **354** may all be parallel. Preferably in this case, one such spring **356** and/or screw **351** is arranged on either side of the adjustment screw means **361** (see below) in a plane parallel to the main connection plane **303**.

(160) It is further noted that such limiting screws **351** may be accessible for adjustment, such as by rotating the screw head **353** in question using a suitable screwdriver or Allen key, from a laterally **302** outer side of the first part **310** and/or the second part **320**. In the preferred case illustrated in FIG. **12**, one such laterally **302** outer side of each of said parts **310**, **320** is arranged with a respective limiting screw channel **354** end hole, via which the limiting screw **351** in question can be reached for rotary adjustment.

(161) As if further illustrated in FIG. **12**, the spring **356** may be arranged around and along said screw **351**, and arranged to be spring-axially compressed between the screw head **353** and a support surface of the first part **310** or of the second part **320**, as the case may be. The support surface may, for instance, be in the form of a shoulder or restriction in said limiting screw channel **354**, limiting a spring-axial freedom of movement of the spring **356** in question.

(162) As mentioned above, the first part **310** may have an abutment surface **313**, and the second part **320** may have an abutment surface **323**, said abutment surfaces **313**, **323** being parallel and possibly perpendicular to the movement dimension **380** but more preferably being parallel to the longitudinal direction **301** (hence set at a non-zero acute angle in relation to the movement dimension **380**). The dimension of movement may advantageously be set at an angle of between 5° and 40°, preferably between 10° and 30°, in relation to said parallel abutment surfaces **313**, **323**.

(163) Then, the abutment surfaces **313**, **323** may be arranged to abut each other when the first part **310** is moved into contact with the second part **320** along said movement direction **380**, reaching said gripping position.

(164) In particular, the spring means **350** may be arranged to press the first part **310** and the second part **320** together, along said movement dimension **380** and possibly under guidance by said guide means, with a force large enough for holding the interface connection adapter arrangement **300** in place along the rifle **2** by friction only (friction between the rifle fastener **331** or **341** and for instance a fastening rail of the rifle **2**), in the absence of any additionally applied gripping force.

(165) As is further illustrated in FIG. **12**, by way of example, the freedom of movement adjustment means **360** may comprise an adjustment screw means **361**, arranged to adjust the freedom of movement of the parts **310**, **320** in relation to each other by screwing the screw means **361** deeper into or out from a thread engagement. The adjustment screw means **261** may in turn comprise a gripping head **362**, arranged externally to said first part **310** and arranged to be translated, by screwing, along the movement direction **380** towards or away from the first part **310**. The gripping head **362** is preferably provided with a high-friction or irregular shape so that a user without difficulty can screw the adjustment screw means **361** by gripping it by the gripping head **362** and applying a screwing force using the user's fingers.

(166) Then, a distance between the gripping head **362** and the first part **310**, such as between an abutment surface of the gripping head **362** (facing a corresponding abutment surface of the first part **310**) and said first part **310** abutment surface, when the first part **310** is far translated towards abutment with the second part **320** as is allowed by a currently gripped rifle fastener rail or similar, defines a maximum freedom of movement from this position, of the first part **310** away from the second part **320** along said movement dimension **380**. In order to achieve this, the adjustment screw means **362** may be fastened, such as using a screw engagement between external threads **366** of the bolt **365** of the adjustment screw means **361** and corresponding internal threads **368** of the channel **367** inside and along which the bolt **365** runs, to the second part **320**.

(167) Alternatively, the gripping head **362** may have internal threads and be arranged to move bolt-axially along bolt **365**, engaging with external threads of bolt **365**, by screwing the gripping head **362** along the bolt **365**.

(168) Hence, the spring means **350** pulls the first part **310** and the second part **320** together into said gripping position. The parts **310**, **320** can be brought apart by applying a manual force, but this will lead to the bolt **365** of the adjustment screw means **362**, being fastened to the second part **320**, moving along said channel **367**, and the gripping head **362** consequently moving towards the first part **310**. When the gripping head **362** reaches abutment with the first part **310**, the first part **310** cannot move further away from the second part **320** along the movement dimension **380** (it is stopped by the gripping head **362**), hence defining a limit to the translational relative freedom of movement of the parts **310**, **320**.

(169) As mentioned above, the gripping head **362** may be rotatable in relation to the bolt axis. By rotating the gripping head **362**, the relative freedom of movement between the parts **310**, **320** is adjusted by the gripping head **362** moving closer to or further from the second part **320**, such as in the movement dimension **380**. This may preferably be accomplished by the gripping head **362** being at least rotationally-fixedly connected to the bolt **365**, so that said thread engagement between threads **366**, **368** is deepened or loosened, as a result axially moving the gripping head **362** in relation to the second part **320**.

(170) Then, the gripping head **362** may further be arranged to, when screwed into abutment with the first part **130** as the first **310** and second **320** parts are in said gripping position, be tightened by further screwing, to secure a desired pressing force between the first part **310** and the second part **320**. Hence, this way the freedom of movement adjustment means **360** may be adjusted to allow a zero freedom of movement, and by further tightening provide a desired tight pressing force effectively fastening the bipod **1** to the rifle **2**.

(171) It is noted that, as used herein, that the parts **310**, **320** are in the "gripping position" may not necessarily imply that the abutment surfaces **313**, **323** abut each other. Depending on the design of the used rifle fastening system, the abutment surfaces **313**, **323** may be separated as the rifle

fastener **331** or **341** is closed into a gripping position where the first **310** and second **320** parts grip about, for instance, a fastening rail of the rifle **2**.

(172) In practise, the channel **367** may comprise a through channel through the first part **310**. Said internal threads **368** may be arranged in the second part **320**, and the gripping head **362** may be arranged on the bolt **365** at an opposite end of the bolt **365** in relation to said external threads **366**.

(173) In some embodiments, the freedom of movement adjustment means **360** further comprises a feedback means **369**, arranged to provide an increased tactile and/or audible feedback to a user screwing the gripping head **362** towards the first part **310**.

(174) In the example illustrated in FIG. **12**, this feedback means **369** comprises a spring-loaded plunger or pin **370** and a set of indentations **364**, the plunger **370** being arranged to interact with the indentations **364** by the plunger **370**, such as a top protrusion **371** of the plunger **370**, being pressed (against the spring-loading force) into several of said indentations **364**, one at a time and in order, as the gripping head **362** is screwed, thereby providing said tactile and/or audible feedback as the plunger moves into and out from each of said indentations **364**.

(175) The fact that the pin **370** is spring-loaded (so that it presses against the underside of the gripping head **362**) also results in a slightly greater pressing force between parts **310**, **320**, leading to a better grip of the bipod in relation to the rifle **2** as the gripping head **362** is tightened.

(176) As an additional effect, the engagement between the indentations **364** and the plunger **370** will prevent the gripping head **362** from accidentally coming undone by being unscrewed, since such unscrewing must overcome a force necessary to move the plunger **370** in/out of engagement with the series of indentations **264** as the gripping head **362** moves in its rotary direction.

(177) As shown in FIG. **12**, the indentations **364** may be arranged in a circular or semi-circular pattern, so that the plunger **370**, describing a circular movement path as the gripping head **362** moves in relation to the first part **310**, follows the pattern formed by the indentations.

(178) FIG. **12** illustrates the plunger **370** being arranged in a plunger channel **372** in the first part **310** and the indentations being arranged in the gripping head **362**, but it may alternatively be the other way around.

(179) Hence, as the gripping head **362** is screwed in, narrowing the translational freedom of movement between parts **310**, **320**, the gripping head **362** moves axially, in the movement dimension **380**, towards the second part **320**. Remember that the first part **310** is held in the gripping position, in relation to the second part **320**, by the spring means **350**. Eventually, the gripping head **362** will move close enough to the first part **310** so that the plunger **370** will come into direct contact with the indentations **364**. As the gripping head **362** is further screwed into deeper threaded engagement of the adjustment screw means **361** with the second part **320**, the plunger **370** will pop further and further into the indentations **364**, providing increasing tactile/audible feedback for each indentation **364** it pops into. As a result, the user can quickly learn to adjust the gripping head **362** to a desired gripping force, irrespectively of a rifle fastening system used, by simply observing the tactile/audible feedback accruing as an integrated part of the fastening of the interface connection adapter arrangement **300** to the rifle **2**.

(180) It is noted that the same freedom of movement adjustment means **360**, and the same gripping head **362** thereof, may be used to operate the bipod **1** (in terms of adjusting a freedom of movement between parts **310**, **320**; to force parts **310**, **320** apart; to fasten the bipod to the rifle by applying additional friction by tightening the screw engagement; etc) irrespectively of if a first rifle fastener **331** or a second rifle fastener **341** is currently used.

(181) As described above, the first part **310** may comprise one or several pairs of fastening holes **312** for fastening an adjustable leg fastening arrangement **200** to the first part **310**, using holes **228** and screws **229**. The second part **320** may comprise corresponding fastening holes **322**.

(182) Both the first part **310** and the second part **320** may be manufactured from metal material, such as steel or aluminium. Preferably, they are each provided as a respective single, integrated metal material body. This provides a sturdy yet light-weight implementation, offering the user

friendliness and flexibility described above.

(183) The various channels and other features of the parts described herein may be manufactured using suitable machining, such as drilling, of a respective integrated metal material body, that may in turn be cast to a desired external shape before such machining takes place.

(184) Description of Methods Related to Said Aspects

(185) FIG. **14** is a flow chart illustrating a first method according to the present invention, for operating a bipod **1** of the present type, the bipod **1** comprising a detachable leg arrangement **100** of the type described herein.

(186) In a first step, the method starts.

(187) In a subsequent step, a detachable leg arrangement **100** of the type described herein is provided. The detachable leg arrangement **100** then comprises the leg part attachment means **120**, in turn comprising the mushroom-shaped support part **123**. For instance, the leg part **110** having the mushroom-shaped support part **123** already mounted thereon, in the way described above, may be provided.

(188) In a subsequent step, the mushroom-shaped support part **123** is introduced into the space **131** of the leg fastening part **130**. This may comprise sliding the mushroom-shaped support part **123** along the space **131**, perpendicularly to the axial direction **111**, to said bottom end of the space **131**.

(189) In a subsequent step, the leg part attachment means **120** is brought into a deepened screw engagement with the leg part **110**, by screwing the leg part **110** in relation to the leg fastening part **120**, until said abutment surfaces **124**, **132** engage, and a flange of the leg fastening part **130** is squeezed between the leg part **110** and the leg part attachment means **120** as described above.

(190) In some embodiments, the detachable leg arrangement **100** is arranged so that the bringing into said deepened screw engagement may comprise turning the leg part **110** less than a full revolution, such as about half a full revolution or less, in relation to the leg part attachment means **120**, such as from a screw position wherein the leg part attachment means **120** is freely movable along said elongated opening **134**, before said abutment surfaces **124**, **132** engage. This can be achieved, for instance, by selecting a thread pitch achieving a sufficient axial translation upon a particular angular turn.

(191) As a result, the leg part **110** is securely fastened to the rest of the bipod **1**, such as to an adjustable leg fastening arrangement **200** of the type described above.

(192) In a subsequent step, the method ends.

(193) FIG. **15** is a flow chart illustrating a second method according to the present invention, for operating a bipod **1** of the present type, the bipod **1** comprising an adjustable leg fastening arrangement **200** of the type described herein.

(194) In a first step, the method starts.

(195) In a subsequent step, said adjustable leg fastening arrangement **200** is provided, being attached to said leg part **110** and to said bipod main part, such as to said interface connection adapter arrangement **300**, or directly to said rifle **2**.

(196) In a subsequent step, the screw engagement means **201** is activated, as described above, so as to achieve said friction engagement having a desired selected leg part **110** pivoting friction, by adjusting a depth of said screw engagement. For instance, this may be accomplished by using an Allen key to turn the pressing screw **226** as described above.

(197) In a subsequent step, said leg part **110** is manually pivoted, while overcoming said friction of said friction engagement, in relation to the bipod main part or the rifle **2**, as the case may be, so as to achieve a desired height and/or orientation of the rifle **2** resting on or supported by the leg part **110**.

(198) In an optional subsequent step, the screw engagement means **201** may be further activated to further increase said friction, effectively fastening the leg part **110** in the current pivotal orientation of the leg part **110**.

(199) It is noted that such a pivotally “fastened” or “locked” leg part **110** may still be possible to

pivot by hand action, as long as the leg part **110** is fastened to the rifle **2**. However, in such a pivotally “fastened” state, the leg part **110** will typically have sufficient pivot friction so as not to pivot by accident during normal use of the rifle in the field. In other words, the adjustable leg fastening arrangement **200** preferably lacks any means to lock the pivoting of the leg part **110**, apart from the pivot-friction engagement described herein, resulting in that pivotally “fastening” means increasing the pivot-friction sufficiently for the leg part **110** to, in practise, be fixed, with respect to a pivoting movement, in relation to the rest of the bipod **1**.

(200) These steps may be repeated any number of times, as is indicated in FIG. **15**, with respect to both leg parts **110** or only one or either of said leg parts **110**. Loosening or fastening the pivoting engagement and moving the leg parts **110** as desired.

(201) Hence, after the leg part **110** in question has been pivoted to a desired orientation, it may again be pivoted to a different desired orientation; the screw engagement means **201** may be further activated to select a different, stronger or weaker, desired pivoting friction; the screw engagement means **201** may be activated to “fasten” the leg part **110**, in the described sense; or the method may end.

(202) After the leg part **110** in question has been “fastened”, as described, the method may end; the screw engagement means **201** may be further activated to set a different, likely slightly weaker, desired pivoting friction; or the screw engagement means **201** may be activated to “loosen” the leg part **110** in question.

(203) Namely, in an optional subsequent step, the screw engagement means **201** may be further activated to loosen the friction, in the sense that it can thereafter be pivoted manually in more easy manner.

(204) Thereafter, in an additional optional step, the leg part(s) **110** may be pivoted to a folding orientation for transport or storage. This step may also be reached directly from the step in which the leg part **110** in question is pivoted to a desired orientation, in case a friction was selected allowing the user to pivot the leg part **110** manually into this folded orientation.

(205) Once in this folded orientation, the screw engagement means **201** may be activated to lock the leg part(s) **110** in the orientation in question, by increasing the friction. This step may be performed while the leg part(s) **110** is/are attached to the interface connection adapter arrangement **300** using the non-parallel hole **312** pairs on either side of the arrangement **300**.

(206) In a subsequent step, the method ends.

(207) FIG. **16** is a flow chart illustrating a third method according to the present invention, for operating a bipod **1** of the present type, the bipod **1** comprising an interface connection adapter arrangement **300** of the type described herein.

(208) In a first step, the method starts.

(209) In a subsequent step, said interface connection adapter arrangement **300** is provided, and in particular being attached to each of said leg parts **110** in a respective orientation in which an axis **111** of the leg part **110** in question between the pivot point of the leg part **110** in question and a distal end **112** of the leg part in question **110** extends on a first side of said main connection plane **303**, as explained above.

(210) In a subsequent step, the interface connection adapter arrangement **300** is connected to a rifle **2**, such as to a fastening rail of the rifle **2**, using the first rifle fastener **331** as described above, such as by moving the parts **310**, **320** into said gripping orientation in relation to the first rifle fastening system of the rifle **2**.

(211) In a subsequent step, said first rifle fastener **331** is detached from the rifle **2**.

(212) In a subsequent step, each of said leg parts **110** is pivoted to a respective orientation in which an axis **111** of the leg part **110** in question between the pivot point of the leg part **110** in question and a distal end **112** of the leg part in question **110** extends on a second side of said main connection plane **303**, as also described above. In practise, each leg part **110** may be pivoted about 180° from its original position.

(213) In a subsequent step, the interface connection adapter arrangement **300** is connected to the same or a different rifle **2**, using the above-described second rifle fastener **341**, such as by moving the parts **310**, **320** into said gripping orientation in relation to the second rifle fastening system of the rifle **2**.

(214) In a subsequent step, the method ends.

(215) FIG. **17** is a flow chart illustrating a fourth method according to the present invention, for operating a bipod **1** of the present type, the bipod **1** comprising an interface connection adapter arrangement **300** of the type described herein.

(216) In a first step, the method starts.

(217) In a subsequent step, said interface connection adapter arrangement **300** is provided, the freedom of movement adjustment means **360** of arrangement **300** being set to define a first freedom of movement of the first part **310** in relation to the second part **320** in said movement direction **380**.

(218) In a subsequent step, the first part **310** is separated from the second part **320**, overcoming a force applied by said spring means **350**, and connecting the connection adapter arrangement **300** to a rifle **2** using said first rifle fastener **331**.

(219) In a subsequent step, the interface connection adapter arrangement **300** is slid along the rifle **2**, such as along a rail of a rifle fastener system of the rifle **2**, until it reaches a desired position.

(220) In a subsequent step, the freedom of movement adjustment means **360** is set to zero freedom of movement of the first part **310** in relation to the second part **320** in said movement direction **380**, thereby locking the interface connection adapter arrangement **300** in position in relation to the rifle **2**. This step may also comprise further tightening the freedom of movement adjustment means **360** so as to tighten a friction engagement between the bipod **1** and the rifle **2**, as described above. The user may use said tactile/audible feedback mechanism to determine a proper tightening.

(221) In an optional subsequent step, the freedom of movement adjustment means **360** is set to define a second freedom of movement of the first part **310** in relation to the second part **320** in said movement direction **380**, the second freedom of movement being smaller than the first freedom of movement but larger than zero. Again, the tactile/audible feedback mechanism to determine a proper tightening being looser than the previously used one.

(222) In a further optional subsequent step, the interface connection adapter arrangement **300** is slid along the rifle **2**, such as along said rail, until it reaches a new desired position. This sliding may take place against the gripping friction provided by the spring means **350**, or by first separating the first part **310** from the second part **320** and then slide the interface connection adapter arrangement **300** without having to overcome said friction.

(223) In a further optional subsequent step, the freedom of movement adjustment means **360** is again set to zero freedom of movement of the first part **310** in relation to the second part **320** in said movement direction **380**, thereby locking the interface connection adapter arrangement **300** in the new desired position in relation to the rifle **2**.

(224) Said sliding steps may comprise a user, using one single hand, pressing the gripping head **362** of the adjustment screw means **361** of said freedom of adjustment means **360** towards the first part **310**, thereby forcing the first part **310** and the second part **320** apart. This may, for instance, be achieved by the user pressing the gripping head **362** (such as using the thumb) while holding (with the rest of the hand) the first part **310** or a part being fastened to the first part, such as the leg part **110** pivotally attached to the first part **310**.

(225) In a subsequent step, the method ends.

(226) Naturally, the first, second, third and/or fourth methods may constitute component parts of a method for using the bipod **1** with the rifle **2**, the method further comprising using the rifle **2** for shooting.

(227) Above, preferred embodiments have been described. However, it is apparent to the skilled person that many modifications can be made to the disclosed embodiments without departing from the basic idea of the invention.

(228) For instance, each of the principles described in connection to the first, second, third and/or fourth aspect of the present invention, and correspondingly the first, second, third and/or fourth method according to the present invention, may be applied independently or in combination with other such aspects/methods, and/or with bipods or methods, as the case may be.

(229) Hence, each of said aspects and methods are freely combinable.

(230) Furthermore, all which has been said about the present bipod **1** is equally applicable to any or all of said methods, and vice versa.

(231) Having said that, the detailed embodiment example presented in the Figures has been selected with the intention of disclosing the various advantages achieved by a bipod **1** and a method according to the present invention, and may of course be varied on a detail level.

(232) Hence, the invention is not limited to the described embodiments, but can be varied within the scope of the enclosed claims.

Claims

1. An interface connection adapter arrangement for a bipod for a rifle, the interface connection adapter arrangement being associated with a main connection plane, wherein the interface connection adapter arrangement is arranged to be pivotally connected to two leg parts, a respective pivot point of both leg parts being arranged in said main connection plane, the leg parts when thus fastened both being arranged to pivot in relation to the interface connection adapter arrangement between a first orientation in which an axis of the leg part in question between said pivot point in question and a distal end of the leg part in question extends on a first side of said main connection plane, and a second orientation in which an axis of the leg part in question between said pivot point in question and a distal end of the leg part in question extends on a second, opposite, side of said main connection plane, wherein the interface connection adapter arrangement comprises: a first interface side, arranged on said first side of the main connection plane, provided with a first rifle fastener, the first rifle fastener being arranged to fasten the interface connection adapter arrangement to the rifle in accordance with a first rifle fastening system, and a second interface side, arranged on said second side of the main connection plane, provided with a second rifle fastener, the second rifle fastener being arranged to fasten the interface connection adapter arrangement to the rifle in accordance with a second rifle fastening system, and wherein the second rifle fastening system is associated with a different fastener geometry as compared to the first rifle connection system; and wherein said interface connection adapter arrangement further comprises a first part and a second part, the first part and the second part being translatable in relation to each other in said main connection plane, wherein said first rifle fastener comprises cooperating fastener means both on said first part and on said second part, wherein said second rifle fastener comprises cooperating fastener means both on said first part and on said second part, and wherein both said first rifle fastener and said second rifle fastener are arranged to be activated for fastening the interface connection adapter arrangement to the rifle by translating said first part in relation to said second part into a gripping position.

2. The interface connection adapter arrangement according to claim 1, wherein said first rifle fastening system is an ARCA system or a Picatinny system, and/or wherein said second rifle fastening system is an ARCA system or a Picatinny system.

3. The interface connection adapter arrangement according to claim 1, wherein said first part and said second part are translatable in relation to each other in a direction having a component parallel to a line interconnecting said pivot points.

4. The interface connection adapter arrangement according to claim 1, wherein the interface connection adapter arrangement comprises an adjustment means for adjusting the relative translational position of the first part in relation to the second part and thereby a gripping force applied by the interface connection adapter arrangement in relation to the rifle, and wherein the

adjustment means is arranged to press the first part and the second part together, thereby achieving a fastening of the interface connection adapter arrangement to the rifle.

5. The interface connection adapter arrangement according to claim 4, wherein said adjustment means is arranged to apply a pressing force of the first part in relation to the second part in a plane being parallel to the main connection plane.

6. The interface connection adapter arrangement according to claim 1, wherein each of said first part and said second part comprises respective fastening means arranged to detachably fasten the first and second part, respectively, to a respective adjustable leg fastening arrangement, each such adjustable leg fastening arrangement being arranged to fasten a respective leg part to the interface connection adapter arrangement.

7. The interface connection adapter arrangement according to claim 6, wherein each of said fastening means is arranged to fasten the adjustable leg fastening arrangement in question to the first or second part in question in at least a first orientation and a second, different, orientation, whereby a centre axis of a pivot cone described by an attached leg part when pivoting in relation to a rifle fastened to the interface connection adapter arrangement is different between said first and second different orientations.

8. The interface connection adapter arrangement according to claim 7, wherein said first orientation allows the leg part in question to be pivoted to an orientation substantially parallel to a shooting axis of the rifle, while said second orientation does not allow the same.

9. A method for operating a bipod comprising an interface connection adapter arrangement according to claim 1, the method comprising the steps: a) providing said interface connection adapter arrangement, being attached to each of said leg parts in a respective orientation in which an axis of the leg part in question between the pivot point of the leg part in question and a distal end of the leg part in question extends on a first side of said main connection plane; b) connecting the interface connection adapter arrangement to a rifle using said first rifle fastener; c) detaching said first rifle fastener; d) pivoting each of said leg parts to a respective orientation in which an axis of the leg part in question between the pivot point of the leg part in question and a distal end of the leg part in question extends on a second side of said main connection plane; and e) connecting the interface connection adapter arrangement to a rifle using said second rifle fastener.
